

EXPERIMENT 5(C) - TIME SERIES ANALYSIS

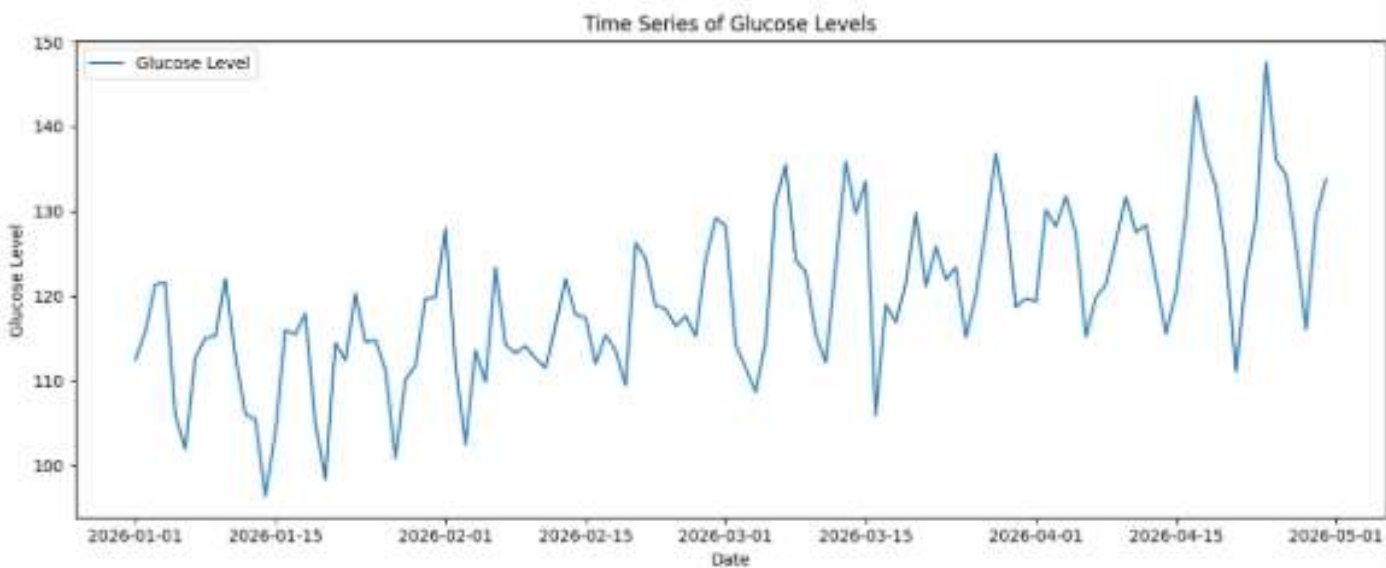
DIABETES TIME SERIES DATA

Glucose

Date	Glucose
2026-01-01	112.48
2026-01-02	115.73
2026-01-03	121.37
2026-01-04	121.59
2026-01-05	106.83

Dataset Shape:
(120, 1)

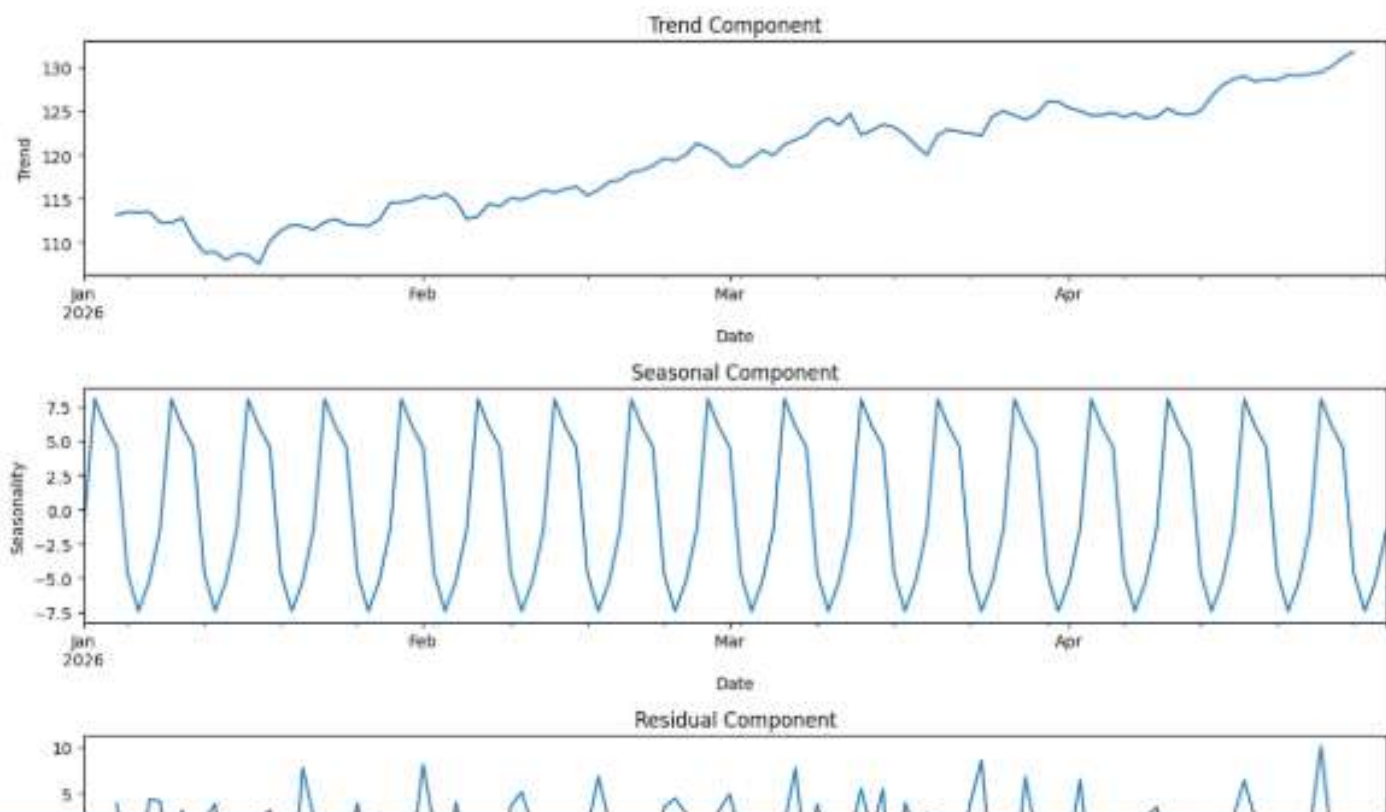
Data Type:
Glucose Float64
dtype: object

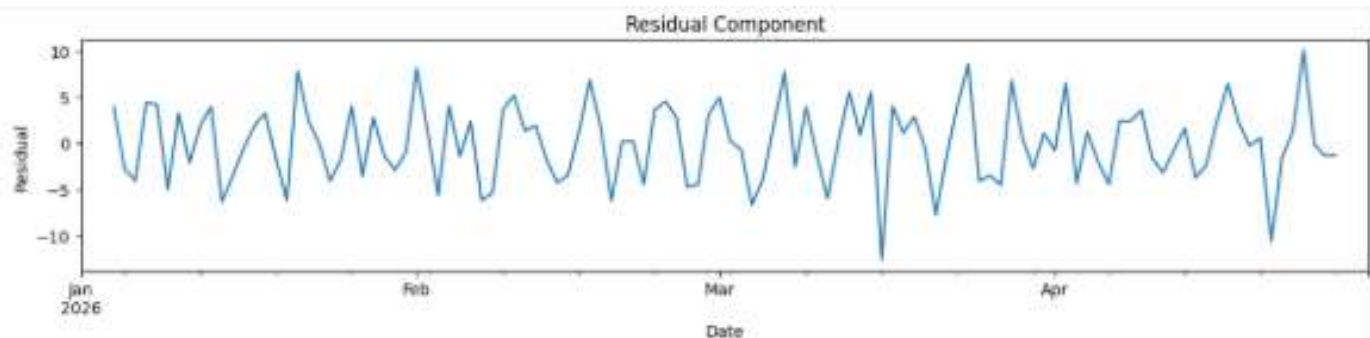


TIME SERIES DECOMPOSITION

Decomposing the glucose time series into:

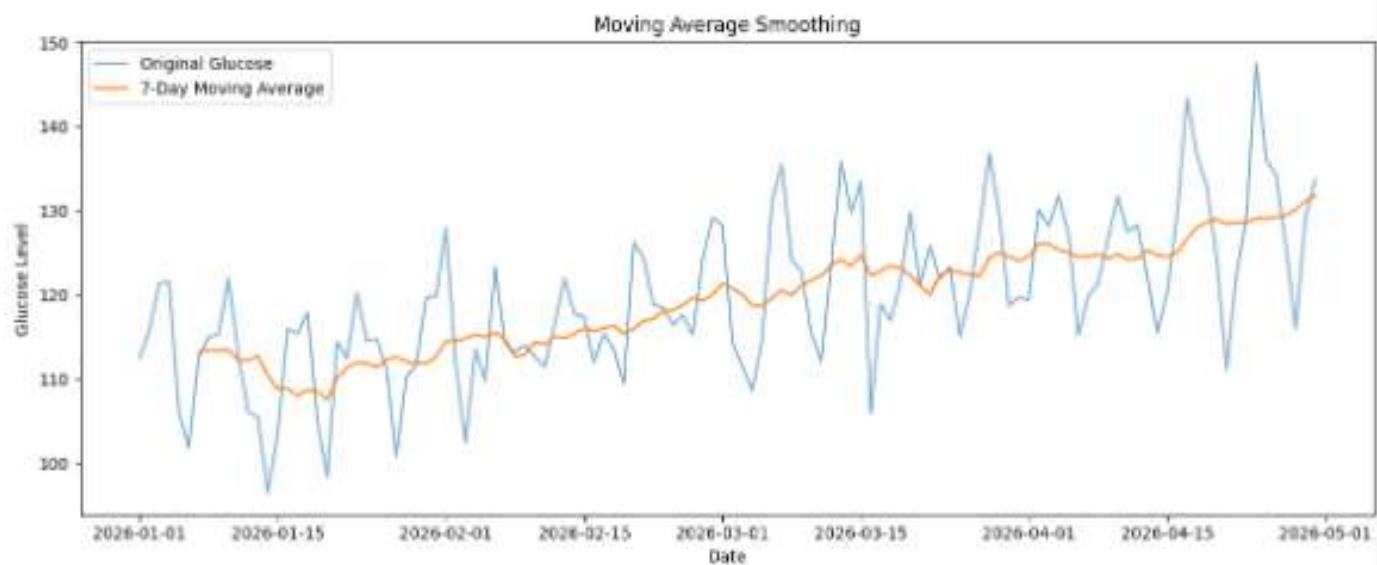
1. Trend
2. Seasonal
3. Residual





7-DAY MOVING AVERAGE

Date	Glucose	Glucose_MA
2026-01-01	112.48	NaN
2026-01-02	115.73	NaN
2026-01-03	121.37	NaN
2026-01-04	121.59	NaN
2026-01-05	105.03	NaN
2026-01-06	101.87	NaN
2026-01-07	112.65	113.102857
2026-01-08	115.01	113.464286
2026-01-09	115.25	113.395714
2026-01-10	122.02	113.488571
2026-01-11	112.83	112.237143
2026-01-12	105.05	112.240000
2026-01-13	105.43	112.748571
2026-01-14	96.36	110.421429
2026-01-15	103.73	108.810000



TRAIN / TEST SPLIT

Total observations : 120
Training observations: 96
Testing observations : 24

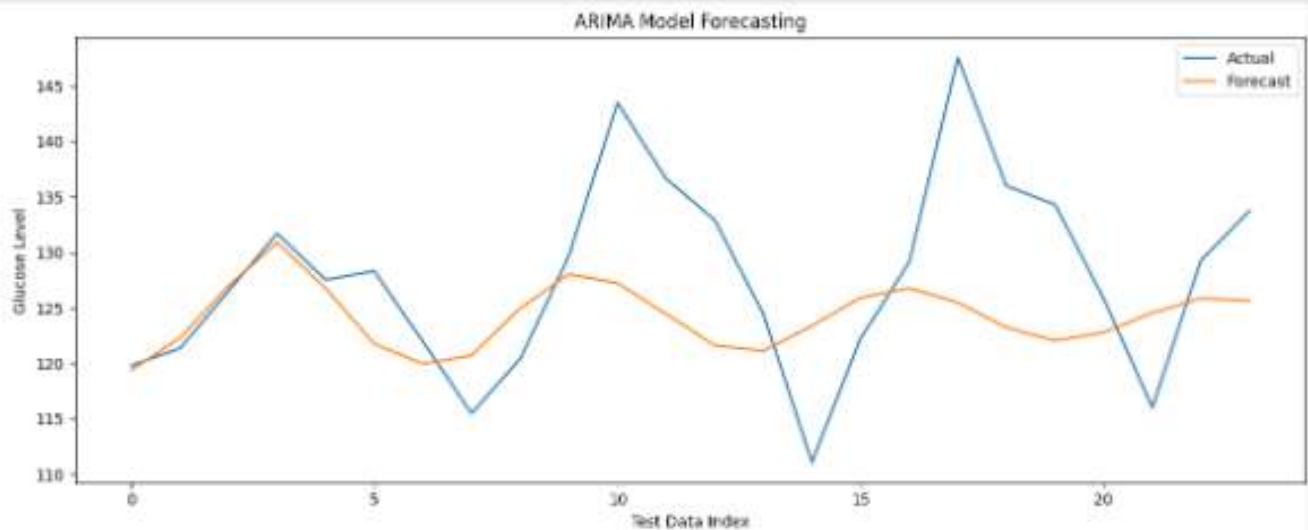
ARIMA MODEL

ARIMA Order: (5, 1, 0)
p = 5
d = 1
q = 0

ARIMA model fitted successfully.

FORECASTED GLUCOSE VALUES

	Actual	Forecast
0	119.82	119.39
1	121.35	122.29
2	126.50	126.97
3	131.72	130.90
4	127.53	126.77



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FORECAST PERFORMANCE

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MAE : 6.4587
MSE : 74.2549
RMSE : 8.6171

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RESULT

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Time Series Analysis was successfully performed on glucose levels.
The original glucose time series was visualized.
Trend, seasonal, and residual components were identified using time-series decomposition.
A 7-day moving average was applied for smoothing.
An ARIMA(5,1,0) model was trained and used to forecast future glucose values.
Actual and forecast values were plotted for comparison.

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EXPERIMENT 5(C) COMPLETED SUCCESSFULLY

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